

# Comparing Large Language Models and Humans in Historical Interpretation

Fabio Celli<sup>1,\*</sup>, Georgios Spathoulas<sup>2</sup>

<sup>1</sup>*Research & Development, Maggioli SpA, Italy*

<sup>2</sup>*Department of Information Security and Communication Technology, NTNU, Norway*

## Abstract

This paper compares how humans and Large Language Models interpret history to perform classification. Using the Structural-Demographic Theory and the Big Cycle as annotation frameworks, we ask three human annotators and three models to annotate the same data using the same instructions, and compare the results. The findings confirm that Large Language Models rival human performance in classification, but interpret history in different ways. While humans tend to disagree on the basis of their personal biases, Large Models disagree by "causal slicing", prioritizing in different ways the facets about multi-causal events through time. We provide recommendations for reducing causal slicing with prompt design.

## Keywords

Large Language Models, Societal Impacts of NLP, Cliodynamics, Cultural Analytics

## 1. Introduction and Related Work

Interpreting historical dynamics is a very difficult task for humans, where subjectivity, prejudice, and Eurocentrism may introduce bias [1]. Historical accounts can be skewed by various influences, such as political motives, cultural beliefs, limited information, and the historian's own viewpoints [2]. In general, bias can be corrected by reaching a consensus or a synthesis between different points of view, or through peer review [3], but cultural biases can be difficult to recognize and overcome [4]. This is true not only for humans, but also for machines. It is well known that algorithms often perpetuate algorithmic bias such as discriminatory inaccuracies rooted in flawed or manipulated data [5].

Large Language Models (LLMs) may also contain biases due to misinformation present in digital data they are trained on. For example the analysis of different models reveals that they frequently reproduce subtle biases regarding minorities in Vietnam, Myanmar, and Nepal [6], which adds to potential cultural hacking issues [7]. Crucially, history is a field where biases can have a huge impact on the interpretation of large scale patterns. Research reveals that LLMs rival human performance in the interpretation of Structural-Demographic phases in history [8, 9], potentially opening new methods for cliodynamics. Understanding the mechanisms that influence historical interpretation is even more important in the age of post-truth, as there is an increasing amount of fake news and misinformation in the media that leads to increased polarization of opinions, potentially both in humans [10] and models [11].

Previous attempts to compare human and Artificial Intelligence in historical interpretations contributed to define the experimental settings for this task by evaluating the response of LLMs against human-annotated data [12, 13] and found that agreement between LLMs depend on the model size [14].

In this paper, we address the issue of historical interpretation by comparing humans and LLMs in the annotation of historical labels (i.e.: growth, apex, decline, crisis) from short descriptions of historical events. We contribute to current research on history interpretation with a quantitative experimental design. In particular, our aim is to answer the following Research Questions:

- (RQ1) How do human raters and LLMs perform in the annotation of historical labels?

---

CLiC-it 2026: Twelfth Italian Conference on Computational Linguistics, September 14–16, 2026, Palermo, Italy

\*Corresponding author.

✉ fabio.celli@maggioli.it (F. Celli); georgios.spathoulas@ntnu.no (G. Spathoulas)

🌐 <https://huggingface.co/facells> (F. Celli); <https://www.ntnu.edu/employees/georgios.spathoulas> (G. Spathoulas)

🆔 0000-0002-7309-5886 (F. Celli); 0000-0003-2947-486X (G. Spathoulas)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

- (RQ2) What types of errors do human raters and LLMs make when interpreting historical data?
- (RQ3) What did humans and machines find most difficult to annotate?

The paper is structured as follows: in Section 2 we provide some background with theoretical frameworks and data. In Section 3 we describe the experimental design, with annotation guidelines, prompts, and the results of the experiments. Finally, in Section 4, we draw our conclusion.

## 2. Annotation Frameworks and Labels

There are many approaches to the study of history, but in general terms they can be grouped into two broad and fuzzy macro-categories. On the one hand, there is cultural/social history that includes authors like Foucault [15] and Harari [16]; on the other hand, there is economic history, including authors like Marx [17] and Piketty [18]. Cultural/social history examines the development and evolution of cultures through time, including artifacts, institutions, ideologies, and revolutions; economic history investigates economic structures and processes throughout history, including production, trade, finance, and technological change.

In our experiment design, we selected one set of labels for cultural/social history, and one for economic history, in order to evaluate the annotation of LLMs and humans with quantitative methods on both the macro-categories. On the side of cultural/social history, we selected the set of labels from the Structural-Demographic Theory (SDT), a framework to understand the dynamics of social and political instability in complex societies [19]. SDT has been applied to a wide range of historical cases, including the French Revolution, the American Civil War [20], the fall of the Qing Dynasty [21], as well as the instability in the US in recent years [22].

**Table 1**  
Phases of the secular cycle (SDT labels)

| Label | Phase                         | Description                                                                                      | Examples                         |
|-------|-------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------|
| 0     | Crisis, Collapse, or Recovery | Elite-led reforms or overthrow, new social equilibrium.                                          | France in 1790s, China in 1950s  |
| 1     | Growth                        | Dynamic period with strong culture, rapid growth, expanding state control.                       | Post-war Italy (1950s)           |
| 2     | Population Impoverishment     | Demographic growth, economic slowdown, wage depression, elite enrichment, social unrest.         | United States in 1890s and 1970s |
| 3     | Elite Overproduction          | Increased access to elite ranks, overloaded social mobility, declining problem-solving capacity. | USSR in 1950s                    |
| 4     | State Stress                  | Weakened governance, elite fragmentation, eroded cooperation, potential violence.                | Germany in the 1920s             |

SDT posits that social unrest arises from the interplay of three actors: the State, with its capacity to maintain order and provide public goods; the elites with their wealth, knowledge, power, and internal divisions; and finally the general population with its demographics and cultures. These actors influence

each other in a cyclical manner through feedback loops. For example, a growing population can strain state resources, leading to increased taxation and elite competition, which can further exacerbate social unrest. Interestingly, unrest is considered a quantitative feature, measurable by the number of rebellions, protests, and revolutions events in a society. Using scientific methods and mathematical modeling, it is possible to identify regularities in the rise and fall of polities over time. Societies typically last one or more secular cycles of variable length [23]. Secular cycles have different phases, gradually increasing social and political instability. Hoyer [24] proposed five phases for the secular cycle, reported in Table 1:

On the side of economic history, we selected the Big Cycle model [25], which identifies three phases from the analysis of the rise and fall of empires in the last 500 years from an economic point of view. The three phases are reported in Table 2:

**Table 2**  
Phases of the Big cycle.

| Label | Phase   | Description                                                                                                  | Examples                      |
|-------|---------|--------------------------------------------------------------------------------------------------------------|-------------------------------|
| 0     | Rise    | A strong society leverages over its competitive advantage and creates peace and prosperity                   | Dutch golden age around 1600s |
| 1     | Apex    | A society must expend increasing resources with reduced returns to maintain its prosperity                   | US around the 2000s           |
| 2     | Decline | A society can no longer meet the costs of maintaining its prosperity and other societies challenge its power | British Empire in the 1940s   |

Big Cycles are driven by the interplay of debt, internal and external instability. As economies expand, credit and debt levels increase, and the distribution of wealth becomes unequal, so productivity decreases with negative impact on economic stability and social cohesion, triggering internal and external instability until the economic cycle restarts. The rise phase roughly corresponds to the growth phase, apex corresponds to the population impoverishment and elite overproduction phases, while decline corresponds to the State stress and crisis phases.

The use of two different annotation schemas (5 vs 2 labels) impacts the difficulty of annotation. Annotating with more labels is perceived as more difficult, and allows us to test two different difficulty settings. In our experimental design, we instruct LLMs and humans to annotate these labels from data describing historical periods, as reported in the next Section.

## 2.1. Data

For our experiments we used the Chronos dataset [26]. The Chronos dataset includes short texts describing polities decade-by-decade from prehistory to 2010 CE in 18 sampling points, including Egypt, Iraq, China, Japan, Italy, Greece, India, Indonesia, Mexico, Hawaii and many others. To verify whether the labeling was consistent, the authors validated the annotation with three people, who independently labeled a sample of 93 examples from the data. They report that the initial agreement was low (Fleiss’  $k$  0.206) because a single disagreement has an exponential impact on the rest of the sequential labels. This is due to the temporal nature of labels, which introduces extra constraints than a normal annotation task, and each misalignment in phase labeling between two annotators reverberates through subsequent time-steps, causing the inter-annotator agreement to decrease exponentially [8]. However, after a training session on SDT, and the use of a standard pattern to start with (the sequence

of secular cycle labels 1,1,2,2,3,3,4,4,4,0), the agreement between humans raised to Fleiss’  $k$  0.455 [26]. The range is similar to the one reported in the literature for Dangerous Speech [27], which also suffers from interpretation problems.

We sampled about 100 decades from different polities in different continents and historical periods like in the following examples:

Decade: 1990s, polity: GeEurri, description: Treaties of Maastricht, Amsterdam and Nice (unified EU currency and created EU as a legal entity). Liberalizations under president Chirac in the 1990s strengthened French economy. Italy was fighting mafia and corruption and made reforms towards a majoritarian electoral system.

Decade: 1170s, polity: CnLrJin, description: Emperor Shizong (r. 1161-1189) confiscated unused land from Jurchen landowners and redistributed to the Jurchen farmers but they preferred to lease the work to Chinese farmers and engage in drinking instead. in 1175 paper factory in Hangzhou employed more than a thousand Chinese workers. Integration problems (language and customs).

Decade: -600s, polity: ItRomRg, description: Lucius Tarquinius Priscus waged wars against the Sabines and Etruscans, doubling the size of Rome and bringing great treasures to the city. To accommodate the influx of population, the Aventine and Caelian hills were populated.

Decade: 1960s, polity: CnPrCr\*, description: from 1966 to 1976 the turbulent period of political and social chaos within China known as the Cultural Revolution (cancel capitalist and traditional culture with violence) led to greater economic and educational decline with millions being purged or subjected to either persecution or politicide. 10 millions students were sent in the country as farmers. War with India for borders.

We included contemporary Europe from 1950s to 2010s, People’s Republic of China from 1950s to 2010s, Mexican Republic from 1820s to 2010s, Later Jin Dynasty from 1130s to 1230s, Mongol Empire from 1240s to 1270s, Yuan Dynasty from 1280s to 1360s, Roman Kingdom from 750s BC to 500s BC and Roman Republic from 490s BC to 280s BC. Each data point contains a timestamp, a polity identifier, and a short description of about 300 characters. The data is publicly available at <https://raw.githubusercontent.com/facells/fabio-celli-publications/refs/heads/main/docs/chronos-human-ai.csv>.

### 3. Experiments

The experimental setting is simple: given the same prompt/guidelines to 3 LLMs and 3 humans, we asked them to annotate the selected examples, providing the label and an explanation of the decision for the two tasks: annotation of Secular Cycle (5 labels task) and Big Cycle (3 labels task).

We randomly selected three LLMs out of a list of twelve available models including Llama, Gemini, GPT, Deepseek, Qwen, Gemma, Mistral and Claude. The selected models are: Claude 3.5 Sonnet (175 billion parameters), Gemini 1.5 Flash (number of parameters not disclosed) and GPT-4 (1.8 trillion parameters). Figure 1 illustrates the experimental workflow.

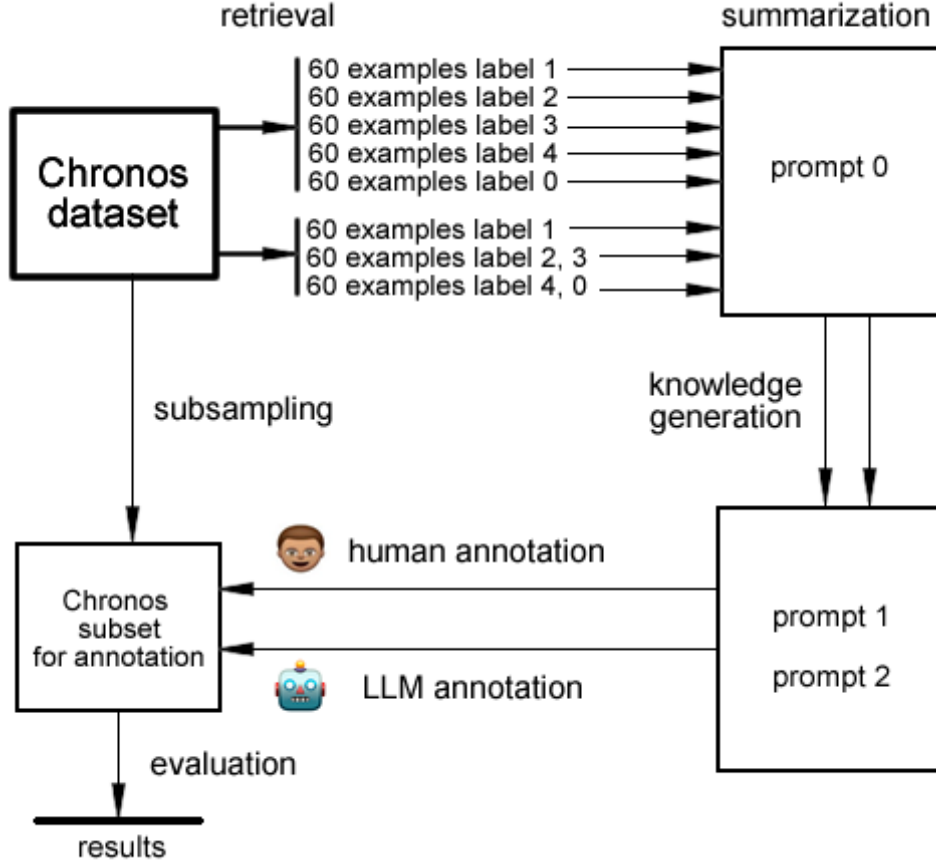
**Table 3**

Prompt 0: summarization and knowledge extraction of cues related to labels.

|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| <p>PROMPT 0<br/>summarize the following phases into prompt instructions optimized to classify the phases, reducing potential overlaps</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------|

In order to produce phase descriptions for the guidelines, we extracted from the Chronos dataset 60 examples of historical decades descriptions per SDT label (reported in Table 1) and summarized them with prompt 0, reported in Table 3.

Then, we used knowledge generation prompting [28] to extract cues associated with each label, and we inserted them in prompts 1 and 2, reported in Tables 4 and 5. In addition to the knowledge generation



**Figure 1:** Schema of the method adopted.

prompting, we also used role-play prompting [29] ("Act as an expert historian"), and few-shot prompting ("Here is an example of the desired output: ..."). These prompts are the guidelines for the annotation of SDT and big cycle phases, respectively, and are used both by LLMs and humans.

We processed the data in bunches from the same periods, in order to let annotators, humans and machines, to have more information about the cycles. We conducted two trials, one with the 5 SDT labels and one with the 3 big cycle labels. Human annotators were students aged 20-25, with no specific background in history. Two different triplets of human annotators were employed in the two trials, while the same LLMs were used for both tasks. We evaluated performance using two metrics: Fleiss'  $k$ , which measures overall agreement, and the average correlation coefficient, which quantifies the similarity in trends across the annotated cycles.

In order to answer RQ1 (how human raters and LLMs perform in the annotation?), we compared the inter-annotator agreement and averaged correlation coefficient among 3 humans and 3 LLMs. The results, reported in Table 6, show that LLMs demonstrate significantly better agreement on historical annotations than humans when interpreting the 5 SDT labels.

However, the picture is different for the 3 labels of the big cycle. In this case, human and machine agreement is comparable, with humans achieving a slightly higher level (0.253). However, even here, LLMs exhibit a higher average correlation coefficient. This suggests that machine-generated annotations tend to identify longer uninterrupted periods, whereas human annotations reflect more fragmented agreement across decades. The agreement between human annotators in the 5 labels task obtained here (0.091) is much lower than Fleiss'  $k=0.206$  reported in previous literature [26]. This suggests that there is more variability than previously reported in the annotation of historical labels, depending on the background culture of the annotators and the clarity of the guidelines. Given this variability, we decided to replicate the experiment with our annotators using the conventional sequence schema reported in

**Table 4**

Prompt 1: annotation of historical data with the 5 labels of the secular cycle. Knowledge generated with prompt 0 is marked in italics.

**PROMPT 1**

Act as an expert historian and consider the Structural Demographic Theory (SDT). Given a set of description of historical decades for different polities, label each description with one of the following secular cycle phases, and provide a short explanation of the choice:

*0=crisis (in this phase may happen societal collapse patterns, power transitions, conflicts, administrative changes, economic patterns, social structure changes, and external influences. Look for signs of civil wars, military coups, environmental factors, population movements, reform of tax systems, trade network disruptions, class conflicts, and foreign invasions).*

*1=growth (a society recovers from a crisis finding a new refresh culture that creates social cohesion. to recognize this phase examine the power structure patterns, legitimacy of rule, social organization, economic patterns, cultural elements, military aspects, and social changes. Look for the presence of elite classes, religious legitimization of power, centralized administrative systems, trade networks, cultural practices, territorial expansion, and population movements.).*

*2=population immiseration (growth slows and inequalities begin to emerge. to recognize this phase evaluate the power dynamics, economic patterns, military aspects, cultural/religious elements, administrative features, and infrastructure development. Look for succession struggles, trade route development, territorial conquests, religious tolerance, bureaucratic reforms, and construction projects.).*

*3=elite overproduction (the number elite aspirants rises, the social lift mechanisms deteriorate. to recognize this phase assess power dynamics, governance, economic patterns, social structures, crisis patterns, cultural and technological development, and common catalysts for change. Look for power struggles, trade system developments, elite class formation, military defeats, religious developments, and military conflicts),*

*4=state stress (elites struggle to institutionalize their advantages. to recognize this phase review political instability, power struggles, economic challenges, military conflicts, administrative changes, and social/religious tensions. Look for succession disputes, financial crises, territorial expansions, bureaucratic reforms, and religious conflicts.)*

A cycle cannot turn back and cannot skip phases. So if in 1940 there is a phase 0, in 1950 there should be a phase 1, in 1960 there can be a phase 1 or phase 2. If in 1960 there is a phase 2, in 1970 there can be a phase 2 or phase 3, not a phase 4. If in 1970 there is a phase 3, in 1980 there can be a phase 3 or 4, and if in 2000 there is phase 4, in 2010 there can be a phase 0 or another phase 4. The decade after phase 0 the cycle restarts from phase 1.

This is an example of the desired output:

decade,polityid,label,explanation

1940,UsUsar\*,0,"(crisis) - war against Nazi Germany. Victory in WW2"

1950,UsUsar\*,1,"(growth) - Expansion to Europe with Marshall Plan, economic growth, civil right movements"

1960,UsUsar\*,1,"(growth) - Continued strong growth, space race, Cuban missile crisis"

1970,UsUsar\*,2,"(immiseration) - Oil crisis, social tensions, inflation"

1980,UsUsar\*,2,"(immiseration) - Recession, welfare retrenchment"

1990,UsUsar\*,3,"(elite overproduction) - Invasion of Kuwait, power consolidation"

2000,UsUsar\*,3,"(elite overproduction) - Terrorism, economic bubbles, financial crisis"

2010,UsUsar\*,4,"(state stress) - Debt crisis, misinformation, social unrest"

literature that obtained  $k=0.455$  with 3 human annotators on 93 examples. We added this sentence to the prompt and run the experiment also with the LLMs: "Initially assume the sequence of labels follows the one of the standard secular cycle model: 1,1,2,2,3,3,4,4,4,0 and then evaluate whether to keep or change the labels in each decade. It is possible to have longer or shorter cycles". The results, reported in Table 7, reveal that LLMs also beat human annotators with this interpretive schema, although the agreement score decreases. It is interesting to note that using the conventional sequence schema, the Fleiss'  $k$  increased for humans by  $\approx 0.25$  points, and this is consistent with the results reported in the literature. Again, this confirms the variability of human annotation, that may depend on personal biases, like opinions and political views. Potentially, using like-minded human annotators it is possible to reach moderate agreement, but when humans have different views, the agreement score drops to slight or no agreement.

Anyway, the effect of the conventional sequence schema on human annotation is to dramatically increase the average correlation coefficient, meaning that there are longer, uninterrupted decades annotated in the same way. The use of this schema with LLMs improves only marginally the average correlation, at the cost of producing more sequence errors, thus decreasing Fleiss'  $k$ . We ran another experiment to answer RQ2 (What types of errors do humans and LLMs make when interpreting historical data?). First of all, we counted the errors due to constraint violations (on the results of the original experiment with 106 examples). Prompt 1 does not allow multiple crisis phases, and both prompts do not allow for phase skipping and backward phases. The results, reported in Table 8, show that LLMs are more prone to skip phases, while humans did not respect the constraint of the two adjacent crisis phases.

In particular, Gemini tends to skip phases in the 5 labels task, but not in the 3 labels task. GPT tends



**Table 5**

Prompt 2: annotation of historical data with the 3 labels of the Big Cycle. Knowledge generated with prompt 0 is marked in *italics*.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <p>PROMPT 2</p> <p>Act as an expert historian and consider the economic cycles. Given a set of description of historical decades for different polities, label each description with one of the following big cycle phases, and provide a short explanation of the choice:</p> <p><i>0=rise (a society recovers from a crisis finding a new refresh culture that creates social cohesion. to recognize this phase examine the power structure patterns, legitimacy of rule, social organization, economic patterns, cultural elements, military aspects, and social changes. Look for the presence of elite classes, religious legitimization of power, centralized administrative systems, trade networks, cultural practices, territorial expansion, and population movements.).</i></p> <p><i>1=apex (growth slows and inequalities begin to emerge. the number elite aspirants rises, the social lift mechanisms deteriorate.. to recognize this phase evaluate the power dynamics, economic patterns, military aspects, cultural/religious elements, administrative features, and infrastructure development. Look for succession struggles, trade route development, territorial conquests, religious tolerance, bureaucratic reforms, and construction projects. Also assess power dynamics, governance, economic patterns, social structures, crisis patterns, cultural and technological development, and common catalysts for change. Look for power struggles, trade system developments, elite class formation, military defeats, religious developments, and military conflicts).</i></p> <p><i>2=decline (in this phase may happen societal collapse patterns, power transitions, conflicts, administrative changes, economic patterns, social structure changes, and external influences. Look for signs of civil wars, military coups, environmental factors, population movements, reform of tax systems, trade network disruptions, class conflicts, and foreign invasions. To recognize this phase also review political instability, power struggles, economic challenges, military conflicts, administrative changes, and social/religious tensions. Look for succession disputes, financial crises, territorial expansions, bureaucratic reforms, and religious conflicts.)</i></p> <p>A cycle cannot turn back and cannot skip phases. So if in 1940 there is a phase 0, in 1950 there should be a phase 1, in 1960 there can be a phase 1 or phase 2. If in 1960 there is a phase 2, in 1970 there can be a phase 2 or phase 0. A phase 0 cannot come after a phase 1.</p> <p>Here is an example of the desired output:</p> |  |  |
| decade,polityid,label,explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |
| 1940,UsUsar*,2,"(decline) - war against Nazi Germany. Victory in WW2"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |
| 1950,UsUsar*,0,"(growth) - Expansion to Europe with Marshall Plan, economic growth"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |
| 1960,UsUsar*,0,"(growth) - Continued strong growth, space race, Cuban missile crisis"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |
| 1970,UsUsar*,1,"(apex) - Oil crisis, social tensions, inflation"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |
| 1980,UsUsar*,1,"(apex) - Recession, welfare retrenchment"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
| 1990,UsUsar*,1,"(apex) - Invasion of Kuwait, power consolidation"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |
| 2000,UsUsar*,1,"(apex)- Terrorism, economic bubbles, financial crisis"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
| 2010,UsUsar*,2,"(decline) - Debt crisis, misinformation, social unrest"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |

**Table 6**

Comparison of LLMs and humans in historical annotation. The best results are marked in bold.

| settings                         | Fleiss k     | avg $\rho$   |
|----------------------------------|--------------|--------------|
| 3 LLMs, 5 labels, 106 examples   | <b>0.330</b> | <b>0.344</b> |
| 3 LLMs, 3 labels, 106 examples   | 0.228        | <b>0.419</b> |
| settings                         | Fleiss k     | avg $\rho$   |
| 3 humans, 5 labels, 106 examples | 0.091        | 0.008        |
| 3 humans, 3 labels, 106 examples | <b>0.253</b> | 0.327        |

**Table 7**

Replication of experiment with sequence schema. The best results are marked in bold.

| settings                        | Fleiss k     | avg $\rho$   |
|---------------------------------|--------------|--------------|
| 3 humans, 5 labels, 93 examples | 0.258        | <b>0.405</b> |
| 3 LLMs, 5 labels, 93 examples   | <b>0.274</b> | 0.366        |

to generate backward phases both with 5 and 3 labels. Claude generates fewer errors than other models. It is interesting to note that both humans and machines do not skip phases in the 3 labels task.

To understand what humans and machines found most difficult to annotate (RQ3), we extracted the explanations where the 3 annotators disagreed and analyzed the frequency of words with a python wordcloud package<sup>1</sup>. Results are reported in Figure 2. The Figures reveal that "war", "crisis", "reform"

<sup>1</sup>[https://github.com/amueller/word\\_cloud/tree/main/wordcloud](https://github.com/amueller/word_cloud/tree/main/wordcloud)

**Table 8**  
Analysis of annotation errors.

| settings       | repeated crises | skipping | backward phase |
|----------------|-----------------|----------|----------------|
| llms, 5 labels | 3               | 7        | 2              |
| llms, 3 labels | -               | 0        | 3              |

| settings         | repeated crises | skipping | backward phase |
|------------------|-----------------|----------|----------------|
| humans, 5 labels | 11              | 3        | 2              |
| humans, 3 labels | -               | 0        | 2              |



**Figure 2:** Wordclouds of the explanations on examples where all 3 annotators were in disagreement.

and "politics" are the most frequent words occurring in context of disagreement between human raters in the 5 labels task. In the 3 labels task, "Roman" and "new" are proxies of difficult interpretations for humans. An interview with human annotators revealed that the application of economic cycles to Roman history was particularly challenging because there was a lot of information about wars and few about the economy. In general, human annotators prioritized economic and political information for their decisions, since these are fields they are more exposed to.

Unlike human raters, LLMs show no specific words associated to contexts of disagreement in the 5 labels task, while the "rise" and "decline" labels are the most difficult to annotate for machines in the 3-labels task. A manual analysis of the explanations generated by LLMs revealed that when the LLMs agree, their interpretation is driven by clean, unidirectional macro-historical events like postwar recoveries or oil crises, which map directly to unmistakable economic or resource signals. Conversely, LLMs' alignment fractures when there are multi-faceted cues, such as major structural wars with gain of resources. In these complex cases, models diverge by "causal slicing", anchoring to different facets of the text. In other words, models prioritize different facets of multi-causal events, and different priorities yield to different final decisions. For example, the French intervention in Mexico has been interpreted by Gemini 1.5 as a consolidation of French power (growth phase), as an elite competition by Claude 3.5 Sonnet (elite overproduction phase) and as an exploitation of resources by GPT 4 (population immiseration phase). In a similar manner, the analysis of the annotations for Dalio's Big Cycle reveals



that LLMs agree on clear, unidirectional milestones. Models often fail to interpret a standardized rule for political consolidation, frequently clashing over whether top-heavy authoritarian power represents a State at its peak efficiency or a desperate crackdown signaling late-cycle decay. Ultimately, these structural disagreements stem from differing temporal framing and a lack of prioritized hierarchies when models evaluate a blend of positive and negative macro-historical indicators. From this perspective, prompts for the annotation of historical labels should be designed with explicit rules to evaluate blends of cues.

## 4. Conclusion

The results presented in this paper confirm that LLMs rival human performance in historical interpretation, especially without the use of conventional sequence schemas that mitigates the exponential increase of error over labels with a sequential nature. In particular, LLMs are strong in the 5-labels task while humans perform best in the 3-labels task, which is easier. Of course both humans and LLMs make errors. While humans tend to inject their own subjective views into historical interpretation, prioritizing things like politics and economy in their decisions, LLMs tend to skip information or sequential phases more than humans. Under this perspective, it would be interesting to repeat the experiments with expert historians to check whether they perform better than non-experts. We leave this for future work. Our analysis of annotator behavior reveals that consensus among LLMs is structurally driven by historical events with clean, unidimensional macro-historical signals, such as explicit institutional creations, post-war recoveries, or profound resource crises. Conversely, when LLM alignment fractures, it is rarely due to faulty logic, but rather to the phenomenon of "causal slicing". In other words, models interpret multi-causal historical phases by anchoring to different socio-demographic facets within the same text block. Given the same input data, one model may assign more importance to elite overproduction cues (e.g. bureaucratic expansion) while another may focus more on cues for population immiseration (e.g. definition of status symbol or population unrest) obtaining different decisions. This is a temporal credit assignment problem (or attribution assignment problem): the challenge of determining which specific input signals, cues, or actions are responsible for a particular outcome in a multi-causal, temporal context. Human brain and reinforcement learning systems solve time-delayed credit assignment by propagating rewards backward through time with a mechanism called Temporal Difference Learning (TD learning) [30, 31]. Eligibility traces keep a temporary memory log of recently observed cues or actions. When a reward or penalty occurs, credit is distributed backward to the cues that were recently active, decaying over time. To mitigate causal slicing and improve LLM performance in Historical Phase Recognition, we recommend that researchers 1) establish an explicit priority tie-breaker hierarchy (prompt instructions should explicitly state to assign priority to the cues that are expected according to the previously annotated phase. For example after annotating a population immiseration phase, the models should start prioritize cues of elite dynamics, and so on; 2) provide a chain context (rather than evaluating historical descriptions as standalone, isolated decades, prompts should be designed to feed data by polity). This would help LLMs to determine whether a polity's trajectory is rising, peaking, or falling relative to its recent past.

These findings have significant implications for digital humanities. While historians' expertise remains crucial for public history discussions, AI now provides powerful tools for annotating large datasets with historical phase. This opens doors for more extensive, reproducible, data-driven historical analysis, helping researchers reduce personal bias. Moreover, LLMs and big data can revolutionize digital humanities by creating new educational experiences, like interactive history learning, exploring alternative historical scenarios, and fostering critical thinking about bias. However, we must be aware of the risks. LLMs trained on biased or inaccurate data can spread misinformation and homogenize narratives. To address this, digital humanities and computational history need to collaborate with trustworthy AI experts. We should focus on making AI annotations transparent, using diverse historical datasets, and comparing outputs from different models. AI offers immense potential, but careful implementation is essential to maximize its benefits and minimize its risks.

## Declaration on Generative AI

During the preparation of this work, the authors used GPT-4 and Gemini 3 in order to perform: Grammar and spelling check; Paraphrase and reword; Improve writing style. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

## References

- [1] M. Wintle, *Eurocentrism: History, identity, white man's burden*, Routledge, 2020.
- [2] R. Bhat, *Interpreting the Past: Historical Approach to Research and Writing*, 2023.
- [3] R. M. Bhat, P. Rajan, L. Gamage, Redressing historical bias: Exploring the path to an accurate representation of the past, *Journal of Social Science* 4 (2023) 698–705.
- [4] C. B. McCullagh, Bias in historical description, interpretation, and explanation, *History and Theory* 39 (2000) 39–66.
- [5] M. C. Jackson, Artificial intelligence & algorithmic bias: the issues with technology reflecting history & humans, *j. Bus. & Tech. l.* 16 (2021) 299.
- [6] M.-K. Mak, T. Luo, A framework for evaluating cultural bias and historical misconceptions in llms outputs, *BenchCouncil Transactions on Benchmarks, Standards and Evaluations* (2025) 100235.
- [7] F. Celli, A. Samal, Monitoring historical cultural hacking in large language models, in: *Proceedings of TRUST-AI 2025 - The European Workshop on Trustworthy AI, co-located with the 28th European Conference on Artificial Intelligence (ECAI 2025)*. Bologna, Italy., 2025, pp. 337–345.
- [8] F. Celli, V. Basile, Large language models rival human performance in historical labeling, in: *International Workshop on Annotation of Real-World Data for Artificial Intelligence Systems*, Springer, 2025, pp. 121–128.
- [9] F. Celli, M. Rovera, Benchmarking historical phase recognition from text and events, in: C. Bosco, E. Jezek, M. Polignano, M. Sanguinetti (Eds.), *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)*, *CEUR Workshop Proceedings*, Cagliari, Italy, 2025, pp. 225–232. URL: <https://aclanthology.org/2025.clicit-1.24/>.
- [10] S. Lewandowsky, U. K. Ecker, J. Cook, Beyond misinformation: Understanding and coping with the “post-truth” era, *Journal of applied research in memory and cognition* 6 (2017) 353–369.
- [11] M. Garry, W. M. Chan, J. Foster, L. A. Henkel, Large language models (llms) and the institutionalization of misinformation, *Trends in cognitive sciences* 28 (2024) 1078–1088.
- [12] J. Hauser, D. Kondor, J. Reddish, M. Benam, E. Cioni, F. Villa, J. S. Bennett, D. Hoyer, P. Francois, P. Turchin, et al., Large language models' expert-level global history knowledge benchmark (hist-llm), in: *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.
- [13] F. Celli, How to compress categorical variables to visualize historical dynamics, in: *Proceedings of the 21st Conference on Information and Research science Connecting to Digital and Library science*, 2025, pp. 1–13.
- [14] F. Celli, D. Mingazov, Knowledge extraction from llms for scalable historical data annotation, *Electronics* 13 (2024) 4990.
- [15] M. Foucault, *Archaeology of knowledge*, routledge, 2013.
- [16] Y. N. Harari, *Nexus: A brief history of information networks from the stone age to AI*, Signal, 2024.
- [17] K. Marx, *Capital: A critical analysis of capitalist production*, Humboldt, 1873.
- [18] T. Piketty, *Capital in the twenty-first century: a multidimensional approach to the history of capital and social classes.*, *British Journal of Sociology* 65 (2014).
- [19] J. A. Goldstone, Demographic structural theory: 25 years on, *Cliodynamics* 8 (2017).
- [20] P. Turchin, *A Structural-Demographic Analysis of American History*, Beresta Books Chaplin, 2016.
- [21] G. Orlandi, D. Hoyer, H. Zhao, J. S. Bennett, M. Benam, K. Kohn, P. Turchin, Structural-demographic analysis of the qing dynasty (1644–1912) collapse in china, *Plos one* 18 (2023) e0289748.

- [22] P. Turchin, A. Korotayev, The 2010 structural-demographic forecast for the 2010–2020 decade: A retrospective assessment, *PloS one* 15 (2020).
- [23] P. Turchin, S. A. Nefedov, *Secular cycles*, Princeton University Press, 2009.
- [24] D. Hoyer, J. S. Bennett, H. Whitehouse, P. François, K. Feeney, J. Levine, J. Reddish, D. Davis, P. Turchin, Flattening the curve: Learning the lessons of world history to mitigate societal crises (2022).
- [25] R. Dalio, *Principles for dealing with the changing world order: Why nations succeed or fail*, Simon and Schuster, 2021.
- [26] F. Celli, V. Basile, History repeats: Historical phase recognition from short texts, *Proceedings of the CLIC-it* (2024).
- [27] V. Tonini, S. Frenda, M. A. Stranisci, V. Patti, et al., How do we counter dangerous speech in Italy?, in: *CEUR Workshop Proceedings*, volume 3878, CEUR-WS, 2024, pp. 1–12.
- [28] Y. Ge, H.-T. Yu, C. Lei, X. Liu, A. Jatowt, K.-s. Kim, S. Lynden, A. Matono, Implicit knowledge-augmented prompting for commonsense explanation generation, *Knowledge and Information Systems* (2025) 1–36.
- [29] Z. Han, Z. Wang, Rethinking the role-play prompting in mathematical reasoning tasks, in: *Proceedings of the 1st Workshop on Efficiency, Security, and Generalization of Multimedia Foundation Models*, 2024, pp. 13–17.
- [30] M. S. Bennett, *A brief history of intelligence: evolution, AI, and the five breakthroughs that made our brains*, HarperCollins, 2023.
- [31] R. S. Sutton, Td models: Modeling the world at a mixture of time scales, in: *Machine learning proceedings 1995*, Elsevier, 1995, pp. 531–539.